

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		Structure – high proportion of collagen (1) Function Any one (x1) from smooth to reduce friction (1) {Elastic / compressible} as a shock absorber (1) Hard to keep trachea open (1) Flexible to allow movement of ribcage (1)	1	1		2		
		(ii)		Auto-immune disease / immune system recognises proteins in the joint as antigens / Immune system attacks {bones / cartilage in joints} (1) Any 2 for one mark of: Cartilage / hyaline, bone synovial membrane / tissue	1	1		2		
	(c)	(i)		Any two (x1) from: {Thinner layer / less} compact bone / ORA (1) Spongy bone has become {less dense / have larger holes} / less bone mass (1) Less hydroxy apatite (1)		2		2		
		(ii)		Any three (x1) from: Treatment decreases the fracture rate compared to {group B / control} (1) Treatment does not decrease the fracture rate to the level in {group A / healthy} (1) Treatment increases the bone strength compared to {group B / control} (1) {group C / experimental} bone strength is not as great as {group A / healthy} (1)			3	3		3
		(iii)		Increased number of osteoblasts to rebuild bones (1) Increases bone density / strength (1)			2	2		
				Question 8 total	6	9	5	20	2	3